	(12) $\frac{2\sigma_0\cos\theta}{\rho g R\left(1-\frac{h}{R}tg\alpha\right)}=h\cos\alpha \quad\Rightarrow\quad \frac{h_0}{1-\frac{h}{R}tg\alpha}=h\cos\alpha$	0,5	
3.4	(12)	0,3	**1,0**
		0.5	
		0,2	
3.5	$\sin\alpha>\frac{R}{4h_0}$	0,6	**1,0**
	$\sin\alpha>0,018$	0,4	
4.1	:	1,5	**3,0**
	(15) $2\sigma_0\pi h^2<2\frac{\pi R^2h^2}{6}\rho g$	1,0	
	( ) $R>\sqrt{\frac{6\sigma_0}{\rho g}}=6,6mm$	0,5	
합계			**10,0**

문제 3. 비선형 축전기 (10,0점)

1. **[0.75점]**

. ,

$$I = 0. \tag{1}$$

, $U_0 =$ 5 V

$$C = 0{,}10\ \mu F. \tag{2}$$

$$q = CU_0 = 0.50\ \mu Cl. \tag{3}$$

2. **[0.25점]** .

$U_0 = 10$ V .

$$t = \infty. \tag{4}$$

3. **[3.0점]** q, C,

.

$$U_0 = \frac{q}{C} + IR, \tag{5}$$

$$I = \frac{dq}{dt}. \tag{6}$$

(6) (5)

$$dt = \frac{R}{U_0 - \frac{q}{C(q)}}dq = f(q)dq, \tag{7}$$

$f(q) = R/\left(U_0 - \frac{q}{C(q)}\right)$.

$f(q)$ $C = C(U)$,

.